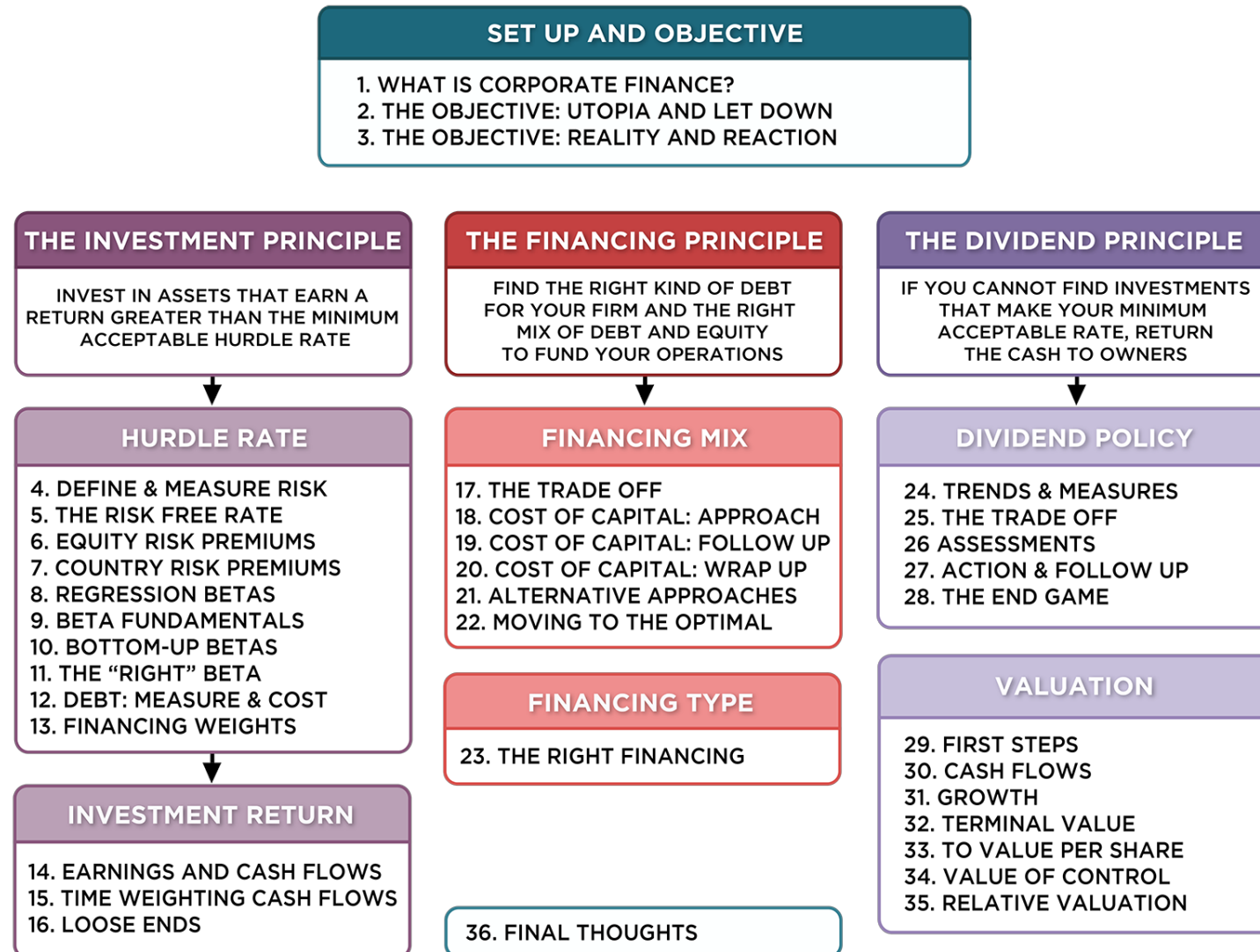


SESSION 06: EQUITY RISK PREMIUMS

Stocks are risky! Really!

SESSION 06: EQUITY RISK PREMIUMS



THE EQUITY RISK PREMIUM

- The risk premium is the premium that investors demand for investing in an average risk investment, relative to the riskfree rate.
- As a general proposition, this premium should be:
 - greater than zero
 - increase with the risk aversion of the investors in that market
 - increase with the riskiness of the “average” risk investment

WHAT IS YOUR RISK PREMIUM?

- Assume that stocks are the only risky assets and that you are offered two investment options:
 - a riskless investment (say a Government Security), on which you can make 3%
 - a mutual fund of all stocks, on which the returns are uncertain
- How much of an expected return would you demand to shift your money from the riskless asset to the mutual fund?

RISK PREMIUMS DO CHANGE . . .

- Go back to the previous example. Assume now that you are making the same choice but that you are making it in the aftermath of a stock market crash (it has dropped 25% in the last month). Would you change your answer?
 - a. I would demand a larger premium
 - b. I would demand a smaller premium
 - c. I would demand the same premium

ESTIMATING RISK PREMIUMS IN PRACTICE

- Survey investors on their desired risk premiums and use the average premium from these surveys.
- Assume that the actual premium delivered over long time periods is equal to the expected premium, i.e., use historical data
- Estimate the implied premium in today's asset prices.

A. THE SURVEY APPROACH

- Surveying all investors in a market place is impractical.
- However, you can survey a few individuals and use these results. In practice, this translates into surveys of the following:

<i>Group Surveyed</i>	<i>Survey done by</i>	<i>Estimated ERP</i>	<i>Notes</i>
Individual Investors	Securities Industries Association	8.3% (2004)	One year premium
Institutional Investors	Merrill Lynch	4.8% (2013)	Monthly updates
CFOs	Campbell Harvey & Graham	4.48% (2012)	5-8% response rate
Analysts	Pablo Fernandez	5.0% (2011)	Lowest standard deviation
Academics	Pable Fernandez	5.7% (2011)	Higher for emerging markets

- The limitations of this approach are:
 - there are no constraints on reasonability (the survey could produce negative risk premiums or risk premiums of 50%)
 - the survey results are extremely volatile
 - they tend to be short term; even the longest surveys do not go beyond one year

B. THE HISTORICAL RISK PREMIUM UNITED STATES – JANUARY 2017

What is the right premium?

1. Go back as far as you can.
Otherwise, the standard error in the estimate will be large.

$$\text{STD ERROR IN ESTIMATE} = \frac{\text{ANNUALIZED STD DEVIATION IN STOCK PRICES}}{\sqrt{\text{NUMBER OF YEARS OF HISTORICAL DATA}}}$$

2. Be consistent in your use of riskfree rate.
3. Use arithmetic premiums for one-year estimates of costs of equity and geometric premiums for estimates of long term costs of equity.

	ARITHMETIC AVERAGE		GEOMETRIC AVERAGE	
	STOCKS - T.BILLS	STOCKS - T.BONDS	STOCKS - T.BILLS	STOCKS - T.BONDS
1928 - 2016	7.96%	6.24%	6.11%	4.62%
STANDARD ERROR	2.13%	2.28%		
1966 - 2016	6.57%	4.37%	5.26%	3.42%
STANDARD ERROR	2.42%	2.74%		
2006 - 2016	7.91%	3.62%	6.15%	2.30%
STANDARD ERROR	6.06%	8.66%		

HISTORICAL RISK PREMIUM

WHAT ABOUT HISTORICAL PREMIUMS FOR OTHER MARKETS?

- Historical data for markets outside the United States is available for much shorter time periods. The problem is even greater in emerging markets.
- The historical premiums that emerge from this data reflects this data problem and there is much greater error associated with the estimates of the premiums.
- Put simply, if you distrust historical risk premiums in the United States, because the estimates are backward looking and noisy, you will trust them even less outside the US, where you have less data.

ONE SOLUTION: BOND DEFAULT SPREADS AS CRP – NOVEMBER 2013

NOVEMBER 2013 - HISTORICAL RISK PREMIUM FOR THE U.S. WAS 4.20%

	ARITHMETIC AVERAGE		GEOMETRIC AVERAGE	
	STOCKS - T.BILLS	STOCKS - T.BONDS	STOCKS - T.BILLS	STOCKS - T.BONDS
1928 - 2012	7.65%	5.88%	5.74%	4.20%
STANDARD ERROR	2.20%	2.33%		

USE THE DEFAULT SPREAD ON THE SOVEREIGN RATING AND
ADD THAT TO THE MATURE MARKET PREMIUM

COUNTRY	MOODY'S RATING	DEFAULT SPREAD	US ERP	TOTAL ERP
BRAZIL	Baa2	2.00%	4.20%	6.20%
CHINA	Aa3	0.80%	4.20%	5.00%
INDIA	Baa3	2.25%	4.20%	6.45%

IF YOU PREFER THE CDS:

COUNTRY	SOVEREIGN CDS SPREAD	US ERP	TOTAL ERP
BRAZIL	2.59%	4.20%	6.79%
CHINA	1.20%	4.20%	5.40%
INDIA	4.20%	4.20%	8.40%

BEYOND THE DEFAULT SPREAD? EQUITIES ARE RISKIER THAN BONDS

- While default risk spreads and equity risk premiums are highly correlated, one would expect equity spreads to be higher than debt spreads. One approach to scaling up the premium is to look at the relative volatility of equities to bonds and to scale up the default spread to reflect this:

$$\text{COUNTRY RISK PREMIUM} = \text{COUNTRY DEFAULT SPREAD} * \left[\frac{\sigma_{\text{EQUITY}}}{\sigma_{\text{COUNTRY BOND}}} \right]$$

- Brazil: The annualized standard deviation in the Brazilian equity index over the previous year is 21 percent, whereas the annualized standard deviation in the Brazilian C-bond is 14 percent.

$$\text{COUNTRY RISK PREMIUM}_{\text{Brazil}} = 4.20\% + 2.00\% * \left[\frac{21\%}{14\%} \right] = 7.20\%$$

- Using the same approach for India and China:

$$\text{COUNTRY RISK PREMIUM}_{\text{India}} = 4.20\% + 2.25\% * \left[\frac{24\%}{17\%} \right] = 7.80\%$$

$$\text{COUNTRY RISK PREMIUM}_{\text{China}} = 4.20\% + 0.80\% * \left[\frac{18\%}{10\%} \right] = 5.64\%$$

Task

Estimate the historical equity risk premium in the market of your choice (if you can)

FOURTH EDITION

Applied Corporate Finance

ASWATH DAMODARAN



WILEY

Optional: Read
Chapter 4